Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0053625053625053	0.9724108658743633 – 1.0274227029389271

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.22349955049075287	0.21900384782544965 – 0.22888203630554305

Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.3114620050231865	0.30519189639903604 – 0.31553349932267266

Thin QR

512 × 128 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.6757057230163083	0.6720568966832867 – 0.6879233649014185

Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.5529138855671462	0.5445555682362697 – 0.556903078560186

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9961673082199184	0.9935358919393151 – 1.0091649948805888

Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0005362505362505	0.9877411308939422 – 1.0212528876206006

Thin QR

256 × 64 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9590911843475237	0.9536949327442831 – 0.9682737393033156

Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0008920606601248	0.9812834224598931 – 1.0268022181146026

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9982135475107777	0.991413560752696 – 1.0239473217976731

Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0035971223021583	0.9870526408502392 – 1.0125065441219243

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0012324316956442	0.9986771773084744 – 1.0135525631939524

Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.001817487237072	0.9901220464195274 – 1.0192704641580272

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.0530631934394596	1.0391581008321096 – 1.0794833581718828

Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0054114825363514	0.9929526907242172 – 1.008696674109766

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.002366793881838	0.9940516705631677 – 1.0146177023358491

Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	0.9773071104387292 – 1.0152439024390243

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	0.9986522911051213 – 1.015047879616963

Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9936608393422213	0.9854601575826497 – 1.009468378111521

Cholesky factorization

256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0032612891221793	0.9983964429344863 – 1.0177890307996873

Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9988555053724515	0.9935237240137919 – 1.0168522189687361

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0118995686103436	0.9831246342890579 – 1.029130116094862

Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0039262481234843	0.987681647914046 – 1.0154321417140948

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0041587746026517	0.9931137845829789 – 1.0251479289940828